Ext

10

Gravitation

In this chapter, we will learn about

- Newton's law of universal gravitation
- finding the gravitational field strength
- the motion of objects in circular orbits in space

Ext

10.1 Universal gravitation

1 Newton's law of universal gravitation

a The gravitational force

Flipped classroom

Watch a video and answer the questions.

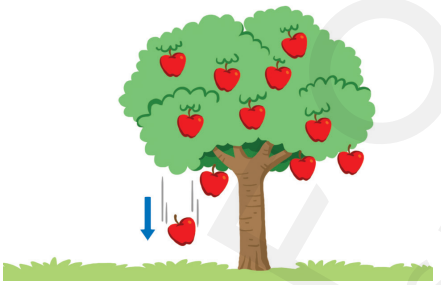
DSE goal

State Newton's law of universal gravitation $F = \frac{Gm_1m_2}{r^2}$.

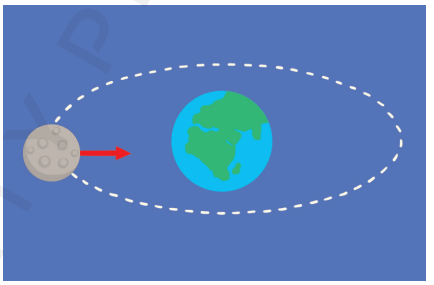
DSE exam

13(1A)Q15, 15(1A)Q11, 17(1A)Q13, 20(1A)Q10, 21(1B)Q5(a)

In the past, most people thought that an apple falling down to the ground and the Moon orbiting the Earth were two different kinds of motion due to different kinds of force (Fig 10.1a). However, Newton thought differently (Fig 10.1b).



(i) An apple falling down to the ground.



(ii) The Moon orbiting the Earth.

Fig 10.1a *Are these motions due to different kinds of force?*

Fig 10.1b *Newton's discovery of universal gravitation.*

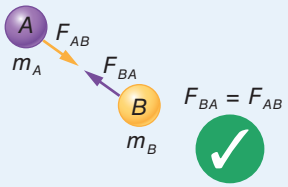
Ext

Right and wrong

Suppose particles *A* and *B* have masses m_A and m_B respectively. If $m_A > m_B$, how will the gravitational forces between *A* and *B* (F_{AB} and F_{BA}) compare with each other?

✗ $F_{BA} > F_{AB}$. Since *A* has a greater mass than *B*, it exerts a larger gravitational force than *B* does.

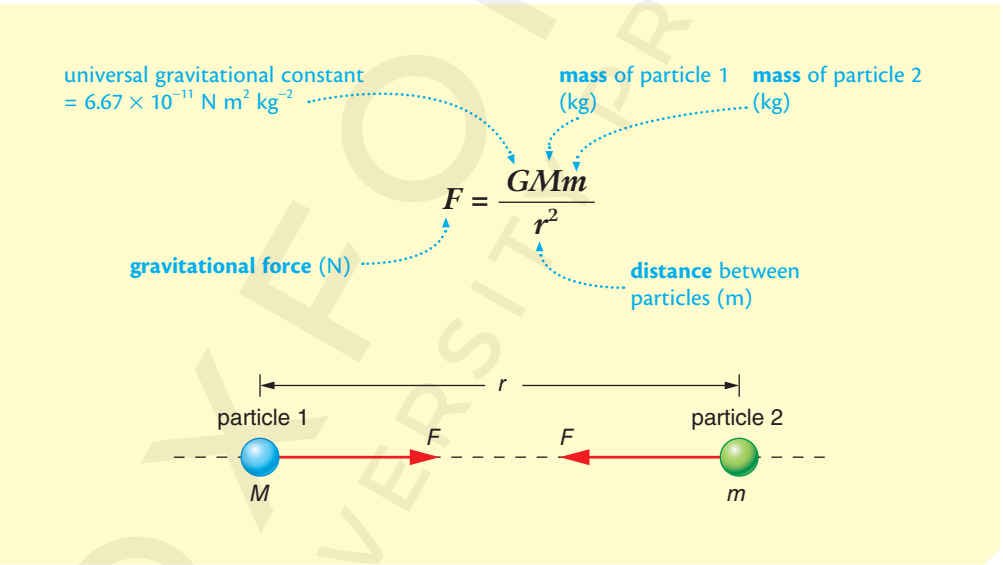
✓ $F_{BA} = F_{AB}$. F_{BA} and F_{AB} are an action-and-reaction pair, so they are equal in magnitude. In fact, the same conclusion can be obtained by applying $F_{BA} = F_{AB} = \frac{Gm_Am_B}{r^2}$.



Newton was the first to realize this: the **gravitational force** which causes an apple to fall to the ground is the same kind of force that keeps the Moon orbiting around the Earth. He went on to conclude that this kind of force exists between all bodies in the universe.

His discovery, now known as universal gravitation, is summarized in **Newton's law of universal gravitation**. The law states that every particle in the universe attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Mathematically, the law can be expressed as:



▶ G is the **universal gravitational constant** and its accepted value is 6.67×10^{-11} N m^2 kg^{-2}.

Rearrange the equation: $G = \frac{Fr^2}{Mm}$.

Unit of $G = \frac{\text{N} \times \text{m} \times \text{m}}{\text{kg} \times \text{kg}}$
$= \text{N m}^2\ \text{kg}^{-2}$

Note that:

1 The gravitational force acts along the line joining the two particles.

2 The gravitational forces acting on the particles form an **action-and-reaction pair**. They are equal in magnitude but opposite in direction.

Know more

The value of *G*

The value of *G* is so small. How did scientists measure this value?

▶ p.445

The equation $F = \frac{GMm}{r^2}$ also applies to objects. The gravitational force can be considered as acting on the centre of gravity (c.g.) of an object and r is taken as the distance between the c.g. of the objects. For a spherically symmetric object, the c.g. is located at the object's centre. See the following example.

gravitational force 引力　Newton's law of universal gravitation 牛頓萬有引力定律
universal gravitational constant 萬有引力常數

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Know more

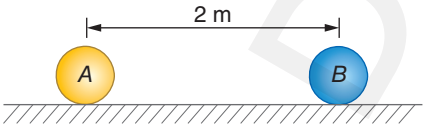
Tides

Tides, the rise and fall of sea levels, are caused by gravitational forces. Do you know how this process works?

▶ p.450

Example 1 Gravitational force acting on objects

Two spherically symmetric objects A and B are placed on the Earth's surface. The distance between their centres is 2 m. The masses of A and B are 7 kg and 5 kg respectively. Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}



(a) Estimate the magnitude of the gravitational force acting on B by A.

(b) Estimate the gravitational force acting on B by the Earth. The Earth's mass is 5.97×10^{24} kg and the Earth's radius is 6370 km.

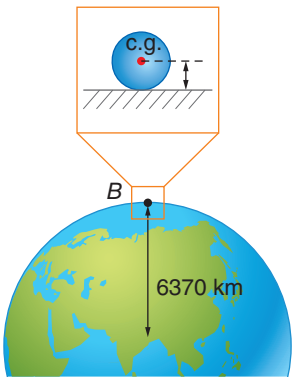
Solution

Tip

r is taken to be the distance between their centres. ▶

(a) Gravitational force $= \dfrac{GMm}{r^2} = \dfrac{(6.67 \times 10^{-11}) \times 7 \times 5}{2^2} = 5.84 \times 10^{-10}$ N

The height of B's c.g. from the ground is too small when compared to the Earth's radius. ▶



(b) For the purpose of estimation, take the distance of B's c.g. from the Earth's centre as 6370 km.

$$F = \frac{GMm}{r^2} = \frac{(6.67 \times 10^{-11}) \times 5 \times (5.97 \times 10^{24})}{(6370 \times 10^3)^2} = 49.1 \text{ N}$$

The gravitational force acting on B by the Earth is 49.1 N (towards the Earth's centre).

Know more

Newton and universal gravitation

It is said that around 1666, Newton was sitting under an apple tree and saw an apple fall to the ground. This inspired him and caused him to formulate the law of gravitation. He attempted to verify it numerically but found that his law was wrong! In 1682, he learned that the value of the Earth's radius that he used in the verification process was incorrect. After Newton used the correct value, the results agreed perfectly with his observations. This law was eventually published in his book *Principia* in 1687.

Statue of Newton looking at an apple.

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Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Complete the table by applying $F = \frac{GMm}{r^2}$. Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

	M	m	r	F
(a)	Sun: 1.99×10^{30} kg	Earth: 5.97×10^{24} kg	1.50×10^{11} m	
(b)	Moon:	Earth: 5.97×10^{24} kg	378 000 km	2.05×10^{20} N
(c)	0.73 kg	158 kg		1.45×10^{-7} N

2 *True or false*: The gravitational force exerted on the Earth by a spaceship is much smaller than that exerted on the spaceship by the Earth. (T/F)

3 When applying $F = \frac{GMm}{r^2}$ to find the force acting on an object of negligible size by the Earth, r is taken as

A the radius of the Earth.

B the distance of the object from the Earth's centre.

C the height of the object above the Earth's surface.

→ Teaching notes and exercises are available on OUP web.

Skill
Graph

Graph of $y \propto \frac{1}{x^2}$

For $y \propto \frac{1}{x^2}$, y decreases when x increases. As x becomes larger, y becomes closer to zero, but never reaches zero. The corresponding graph may be sketched with the help of the following table. Note that x^2y = constant all the time.

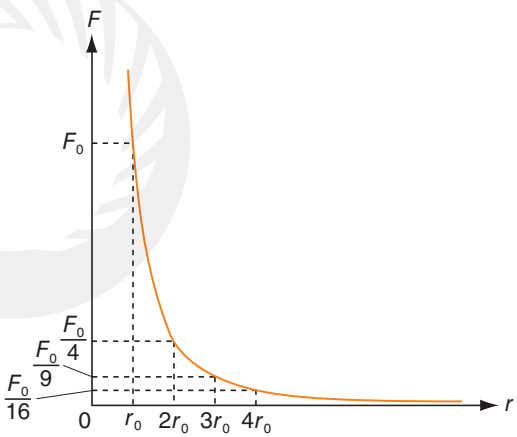
x	y
r_0	F_0
$2r_0$	$\frac{F_0}{4}$
$3r_0$	$\frac{F_0}{9}$
$4r_0$	$\frac{F_0}{16}$

Simulation

→ This simulation allows Ss to vary M and m and see how F changes with r. It is available on OUP web in PC version (.exe) only.

b The inverse-square relationship

According to $F = \frac{GMm}{r^2}$, for any two objects with their c.g. separated by a distance r, the gravitational force F acting on the two objects is inversely proportional to r^2 , i.e. $F \propto \frac{1}{r^2}$. This is an example of the *inverse-square law*. The force weakens quickly as the objects are further apart (Fig 10.1c).



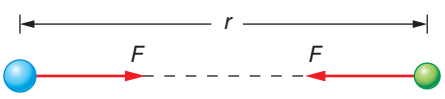


Fig 10.1c *The graph of F against r, showing their inverse-square relationship.*

inverse-square law 平方反比定律

Ext

Know more

Inverse-square law

The inverse-square law can be more easily understood by thinking about spray painting? To learn more, turn to the page indicated below.

▶ p.446

Example 2 Gravitational force, mass and distance

Two particles X and Y are separated by r. The mass of X is m. The gravitational force between X and Y is F.

What happens to F if

(a) m is doubled?

(b) r is doubled?

Solution

(a) Since $F \propto m$,

$$\frac{F_1}{F_0} = \frac{m_1}{m_0} = 2$$

$$\Rightarrow F_1 = 2F_0$$

F will double if m is doubled.

(b) Since $F \propto \frac{1}{r^2}$,

$$\frac{F_1}{F_0} = \frac{\frac{1}{r_1^2}}{\frac{1}{r_0^2}} = \left(\frac{r_0}{r_1}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\Rightarrow F_1 = \frac{1}{4}F_0$$

F will become $\frac{1}{4}$ of the original value if r is doubled.

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Which of the following pairs of quantities obey(s) the inverse-square law? Put a tick in the corresponding box(es).

☐ E and v in $E = \frac{1}{2}mv^2$

☐ I and r in $I = \frac{P}{4\pi r^2}$

☐ F and r in $F = \frac{mv^2}{r}$

☐ F and r in $F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$

Ext

Learn by practice 1

Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

1 Two particles A and B are 2 m apart. The mass of A is 1.5 kg and the mass of B is 3.0 kg.

(a) Find the gravitational force exerted on A by B.

(b) Find the gravitational force exerted on B by A.

2 Two particles each have a mass of 0.2 kg. How far should they be so that the gravitational force acting between them is 3×10^{-9} N?

3 The gravitational force between two particles is F when their distance is r. What is the gravitational force between them when their distance is

(a) $3r$ (b) $0.5r$

2 Mass, weight and acceleration due to gravity

Consider an object falling on the Earth in the absence of air resistance. Due to its weight, the object moves with the acceleration due to gravity g, which is constant and equal to 9.81 m s^{-2}. However, this is **only true when the object is close to the Earth's surface**. If the object is at a great distance from the Earth, g is not 9.81 m s^{-2}.

Here, an object close to the Earth's surface means that its distance from the Earth's surface is very small compared with the Earth's radius.

Suppose a falling object is at a certain distance r from the Earth's centre. The only force that the object experiences is its own weight, i.e. the gravitational force acting on it by the Earth (Fig 10.1d).

$r = R_E + d$

where R_E = Earth's radius, d = distance of object from the Earth's surface

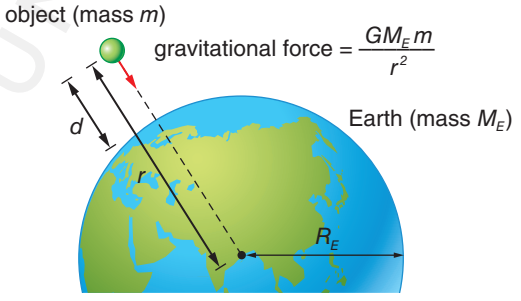


Fig 10.1d *Gravitational force experienced by a falling object at distance r from the Earth's centre.*

By Newton's law of universal gravitation,

$$\text{weight} = \text{gravitational force} = \frac{GM_E m}{r^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

By Newton's second law,

$$\text{weight} = mg \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (2)$$

Ext

Combining equations (1) and (2), we have

$$mg = \frac{GM_E m}{r^2}$$

Therefore, the acceleration due to gravity g from the Earth can be expressed as

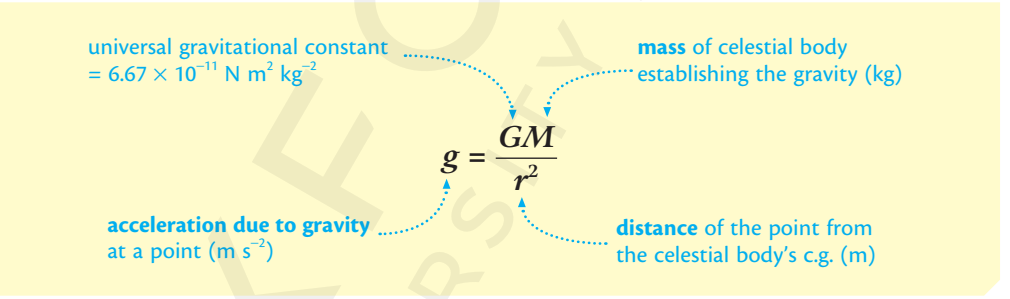
$$g = \frac{GM_E}{r^2}$$

For an object near the Earth's surface, d is small and $r \approx R_E$.

For an object at a great distance from the Earth's surface, d is comparable to R_E and r cannot be taken as the constant R_E.

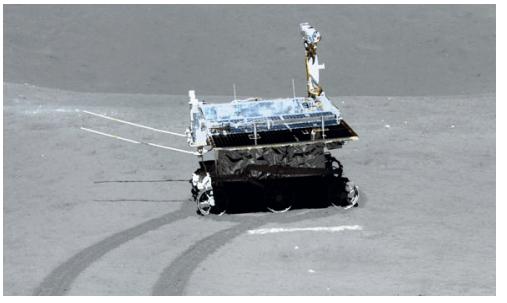
▶ We can see that g depends on r and they obey the inverse square law. g cannot be regarded as constant when the object is at a great distance from the Earth's surface.

The above equation can also be used to express the acceleration due to gravity from other celestial bodies.



Example 3 | Weight on the Moon

Yutu is the first Chinese lunar rover to operate on the Moon. Its mass is 140 kg.



Given: Moon's mass = 7.35×10^{22} kg, Moon's radius = 1.74×10^{6} m

Assume the effect of the Earth and the Sun is negligible. Determine the following quantities as measured on the Moon's surface:

(a) the acceleration due to gravity,

(b) the weight of Yutu rover.

Solution

(a) Acceleration due to gravity $g = \frac{GM}{r^2} = \frac{(6.67 \times 10^{-11}) \times (7.35 \times 10^{22})}{(1.74 \times 10^{6})^2} = 1.62 \text{ m s}^{-2}$

(b) Weight = $mg = 140 \times 1.62 = 227$ N

Yutu 玉兔號

Ext

Google Forms

Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 A spacecraft is approaching the Earth under the effect of Earth's gravity only (Fig a).

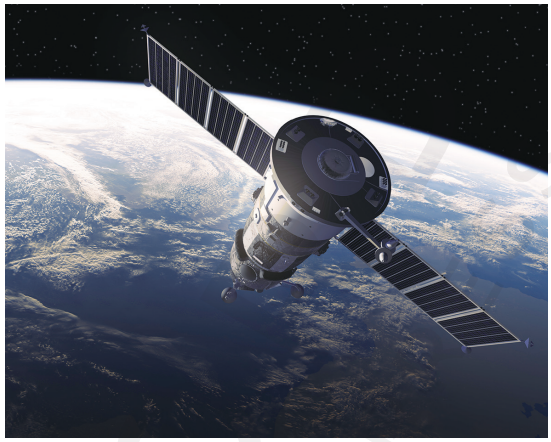


Fig a

At that moment, is each of the following quantities increasing, decreasing or remaining unchanged?

(a) The mass of the spacecraft

(b) The weight of the spacecraft

(c) The acceleration of the spacecraft

Now, consider the acceleration due to gravity on the Earth's surface, g_0. Given that the Earth's mass $M_E = 5.97 \times 10^{24}$ kg and the Earth's radius $R_E = 6.37 \times 10^6$ m, we have:

$$g_0 = \frac{GM_E}{R_E^{\,2}} = \frac{(6.67 \times 10^{-11}) \times (5.97 \times 10^{24})}{(6.37 \times 10^6)^2} = 9.81 \text{ m s}^{-2}$$

This agrees with the value given in chapter 2.4 (p.73). Note that g_0 is a known value; the value of g at any distance r away from the Earth's centre can then be expressed in terms of g_0 as follows:

$$\frac{g}{g_0} = \frac{\dfrac{GM_E}{r^2}}{\dfrac{GM_E}{R_E^{\,2}}} = \frac{R_E^{\,2}}{r^2}$$

$$\Rightarrow g = g_0 \times \frac{R_E^{\,2}}{r^2}$$

The relation is obtained from the ratio of g on the Earth's surface to that at a distance r from the Earth's centre.

Sometimes we may need to solve problems using this relationship, as shown in Examples 4 and 5.

Ext

Example 4 Weight of astronaut

Cooper is an astronaut. His mass is 60 kg. The Earth's radius is R_E. Assume the effect of the Moon and the Sun is negligible. Determine the weight of Cooper when he is

(a) on the Earth's surface;

(b) at $2R_E$ above the Earth's surface during a space mission.

Solution

(a) Weight $= mg_0 = 60 \times 9.81 = 589$ N

Tip

'At $2R_E$ above the Earth's surface' means 'at $3R_E$ from the Earth's centre'.

(b) $$\frac{g}{g_0} = \frac{\frac{GM_E}{r^2}}{\frac{GM_E}{R_E^2}} = \frac{R_E^2}{r^2} = \frac{R_E^2}{(3R_E)^2} = \frac{1}{9}$$

$$\Rightarrow g = 9.81 \times \frac{1}{9} = 1.09 \text{ m s}^{-2}$$

Weight $= mg = 60 \times 1.09 = 65.4$ N

Example 5 Effect of the Sun's gravity

The mass of the Sun is about 3.33×10^5 times that of the Earth. Estimate the acceleration due to gravity of the Sun on the Earth's surface. Given: Earth's radius $R_E = 6.37 \times 10^6$ m, Earth-Sun distance $D_S = 1.50 \times 10^{11}$ m

Solution

$$\frac{g}{g_0} = \frac{\frac{GM_S}{D_S^2}}{\frac{GM_E}{R_E^2}}$$

$$= \left(\frac{M_S}{M_E}\right)\left(\frac{R_E^2}{D_S^2}\right)$$

$$= (3.33 \times 10^5) \times \frac{(6.37 \times 10^6)^2}{(1.50 \times 10^{11})^2}$$

$$= 6.005 \times 10^{-4}$$

$$\Rightarrow g = 9.81 \times (6.005 \times 10^{-4}) = 0.005\ 89 \text{ m s}^{-2}$$

The acceleration due to gravity of the Sun on the Earth's surface is $0.005\ 89$ m s^{-2}.

Ext

Learn by practice 2

1 Consider a rock of mass m. Let its weight be W and its acceleration due to gravity be g at a certain location. The radius of the Earth is R_E. Complete Table a.

Location	m / kg	W / N	g / m s^{-2}
Earth's surface		35.2	9.81
R_E above Earth's surface			
$3R_E$ above Earth's surface			

Table a

2 Find the acceleration due to gravity g on the surfaces of the following planets with different masses and radii. M_E and R_E are the mass and the radius of the Earth respectively.

(a) Planet P of mass $1.2M_E$ and radius R_E

(b) Planet Q of mass M_E and radius $1.5R_E$

(c) Planet R of mass $2M_E$ and radius $3R_E$

DSE goal

Define gravitational field strength as force per unit mass.

Define the gravitational field strength at a point above a planet.

3 Gravitational field

Recall that the electric force and the magnetic force are also non-contact forces.

Gravitational force is a non-contact force—the force can be applied by one object on another **without any contact**. To understand non-contact forces, we use the concept of the *field*.

The gravitational field is a scientific model used to explain the influences of an object in the space around it due to its mass.

According to this concept, an object establishes a **gravitational field** in the space around it. Any other objects in the field will experience a gravitational force (Fig 10.1e). The size of the force depends on the strength of the field.

Suppose there is an imaginary place with no gravitational field. The spacecraft in this place experiences no gravitational force.

Since the Earth establishes a gravitational field around it, any objects within this field experiences a gravitational force.

Fig 10.1e *A spacecraft experiences a gravitational force in a gravitational field.*

field 場　　gravitational field 引力場

Ext

Both the acceleration due to gravity and the gravitational field strength are represented by g, but they have different meanings (see p.427).

▶ Suppose a small test mass m_0 at a certain point in a gravitational field experiences a gravitational force F. The **gravitational field strength**, denoted by g, **at that point** is defined as (Fig 10.1f):

$$g = \frac{F}{m_0}$$

gravitational field strength at a point (N kg^{-1})

gravitational force experienced by the mass at that point (N)

mass (kg)

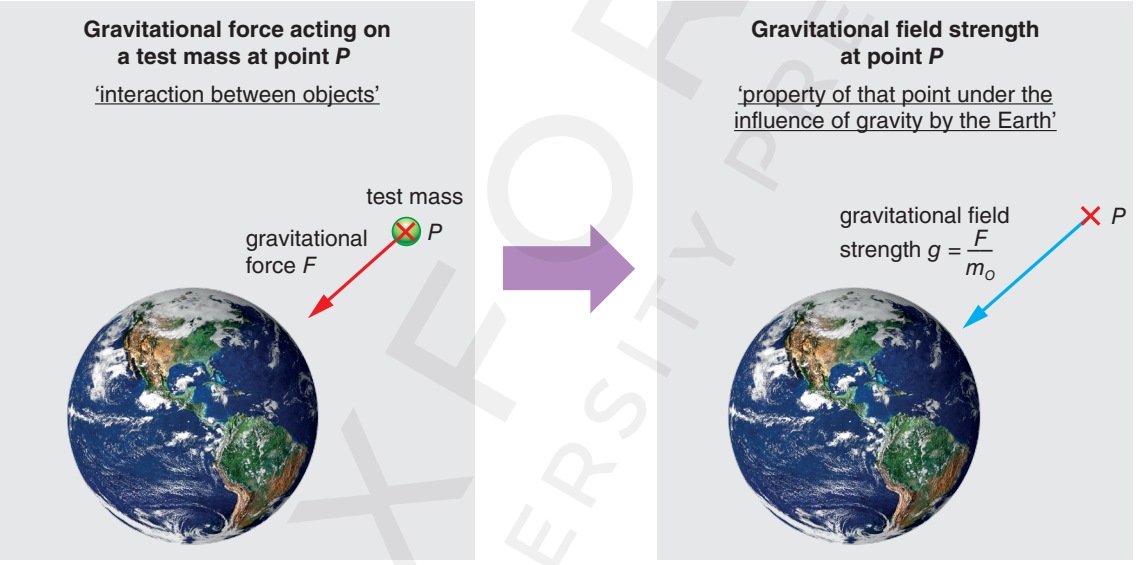


Fig 10.1f *Gravitational force and gravitational field strength.*

The gravitational field strength is measured in **N kg^{-1}**. It is a vector with a direction the same as the gravitational force F acting on the test mass.

We can find gravitational force F on a mass m at any particular point in a gravitational field by

$$F = mg$$

where g is the gravitational field strength at that point.

Suppose an object of mass M establishes a gravitational field around it. According to Newton's law of universal gravitation, the gravitational force F acting on a test mass m_0 at a distance r away from the object's c.g. is

$$F = \frac{GMm_0}{r^2}$$

Gravitational force is a non-contact force. Scientists thought of this force as a direct interaction between two objects. This point of view is called action-at-a-distance, and can be represented by

mass ⇔ mass

However, in this point of view, the effect of moving one object is instantaneously transmitted to another object. This violates special relativity, which limits the transmission speed of such information to the speed of light.

Scientists then adopted the field concept, which regards an object as modifying the space around it and setting up a gravitational field. Any other object in this field is acted by a gravitational force. The field therefore plays an intermediate role:

mass ⇔ field ⇔ mass

gravitational field strength 引力場強度

Ext

Therefore, the **gravitational field strength** g there is:

$$g = \frac{F}{m_0} = \frac{\dfrac{GMm_0}{r^2}}{m_0} = \frac{GM}{r^2}$$

universal gravitational constant = 6.67×10^{-11} N m^2 kg^{-2}

mass of object establishing the gravitational field (kg)

$$g = \frac{GM}{r^2}$$

gravitational field strength at a point (N kg^{-1})

distance of the point from the object's c.g. (m)

The gravitational field strength g is independent of test mass m_0. However, g and r obey the inverse-square law, i.e. $g \propto \frac{1}{r^2}$. Figure 10.1g shows the graph of the gravitational field strength g against r for the Earth.

The gravitational field strength inside the Earth, i.e. when $r < R_E$, lies beyond the scope of this book.

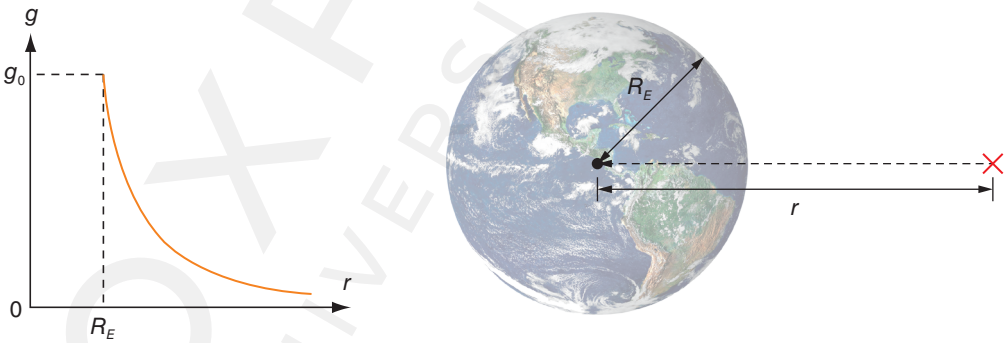


Fig 10.1g *The graph of g against r, showing their inverse-square relationship.*

Example 6 Gravitational field strength above the Earth's surface

(a) Show that the magnitude of the gravitational field strength g_0 due to the Earth at the Earth's surface is 9.81 N kg^{-1}.

Given: Earth's mass $M_E = 5.97 \times 10^{24}$ kg, Earth's radius $R_E = 6.37 \times 10^6$ m

(b) A satellite of mass 4500 kg is at a height of $3R_E$ above the Earth's surface.

(i) Find the magnitude of the gravitational field strength due to the Earth at the position of the satellite.

(ii) Find the magnitude of the gravitational force acting on the satellite by the Earth.

Ext

Solution

(a) $g_0 = \dfrac{GM_E}{R_E^2} = \dfrac{(6.67 \times 10^{-11}) \times (5.97 \times 10^{24})}{(6.37 \times 10^6)^2} = 9.81 \text{ N kg}^{-1}$

(b) (i) $\dfrac{g}{g_0} = \dfrac{\dfrac{GM_E}{r^2}}{\dfrac{GM_E}{R_E^2}} = \dfrac{R_E^2}{r^2} = \dfrac{R_E^2}{(4R_E)^2} = \dfrac{1}{16}$

$\Rightarrow g = 9.81 \times \dfrac{1}{16} = 0.613 \text{ N kg}^{-1}$

The magnitude of the gravitational field strength is 0.613 N kg^{-1}.

(ii) $F = mg = 4500 \times 0.613 = 2760$ N

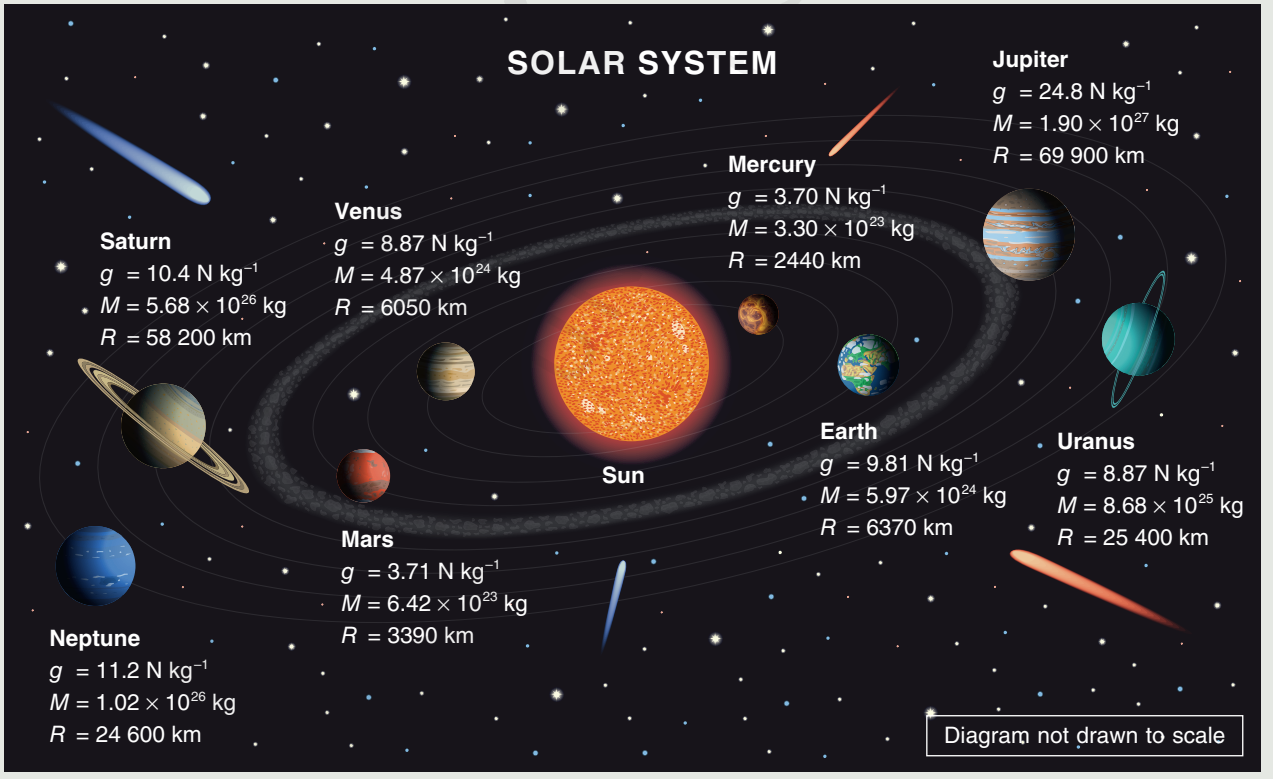
The magnitude of the gravitational force is 2760 N.

Know more

→ This simulation shows how the motion of a person is affected when he jumps on different planets.

Gravitational field strength, mass and radius of major planets in the solar system

Simulation



Ext

4 Meaning of *g*

The symbol g can be used to denote two different but very closely related quantities:

1 the acceleration of an object at a point when it falls freely under gravity;

2 the gravitational field strength at that point.

As both quantities can be expressed as $g = \frac{GM}{r^2}$, they **always have the same numerical value at a given point**. Figure 10.1h shows the case at a point near the Earth's surface.

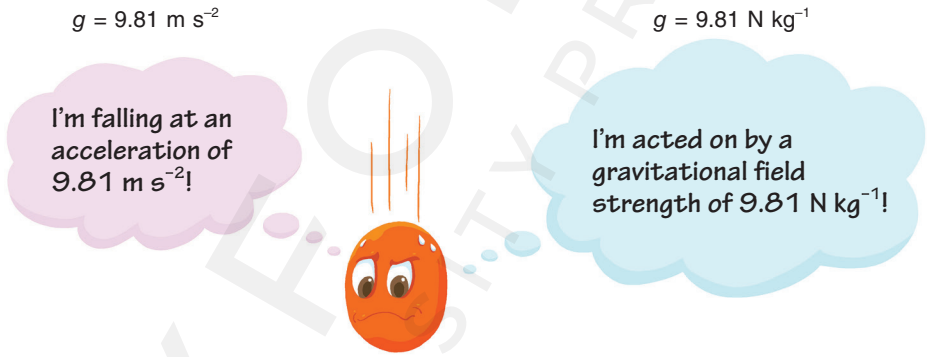


Fig 10.1h *g represents two different quantities whose numerical value are the same at a given point.*

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false*:

(a) The gravitational field strength at a point is the gravitational force per unit mass acting on an object at that point. (T/F)

(b) If two satellites orbit the Earth at the same height above the Earth's surface, the one with a larger mass will experience a greater gravitational field strength from the Earth. (T/F)

(c) If X and Y are two nearby stars, and the gravitational field strength at X due to Y is g_0, then the gravitational field strength at Y due to X must also be g_0. (T/F)

(d) If the radius of the Earth is R_E, and the gravitational field strength on the Earth's surface is g_0, then the gravitational field strength at $2R_E$ above the Earth's surface is $0.25g_0$. (T/F)

2 Consider an isolated system of two point masses X and Y located at P and Q as shown (Fig a). The distance between X and Y is r. The masses of X and Y are m_1 and m_2 respectively.

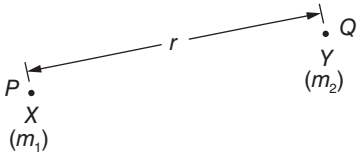


Fig a

Among r, m_1 and m_2, which quantities should be used to find each of (a) to (d) listed below?

(a) Gravitational field strength at P due to Y

(b) Gravitational force acting on X by Y

(c) Gravitational field strength at Q due to X

(d) Gravitational force acting on Y by X

Ext

Exercise 10.1

Take $g = 9.81\ \text{m s}^{-2}$ (close to the Earth)
$G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$

1 Which of the following statements about the acceleration due to gravity g and the universal gravitational constant G is correct?

A The value of g remains unchanged at different positions.

B The value of G remains unchanged at different positions.

C The unit of g is the same as that of G.

D Both g and G are vectors.

2 Objects X and Y are r apart. The mass of X is m_1 and the mass of Y is m_2. What is the gravitational field strength at X due to Y?

A $\frac{Gm_1}{r^2}$ B $\frac{Gm_2}{r^2}$

C $\frac{Gm_1m_2}{r^2}$ D $\frac{Gm_1m_2}{r}$

***3** The mass of the Earth is about 9 times that of Mars, and its distance from the Sun is only $\frac{2}{3}$ of the distance between Mars and the Sun. If the gravitational force acting on Mars by the Sun is F, estimate the gravitational force acting on the Earth by the Sun.

A $4F$ B $6F$

C $14F$ D $20F$

***4** Which of the following statements about the gravitational forces between objects is/are correct?

(1) An object experiences a gravitational force because it is located in the gravitational field developed by its own.

(2) Gravitational forces between objects are always attractive.

(3) The magnitude of a gravitational force is inversely proportional to the distance between objects.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

***5** A rock is moving radially towards a planet (Fig a). Which of the following statements about its motion is correct? Neglect the effect of other celestial bodies and resistive forces.

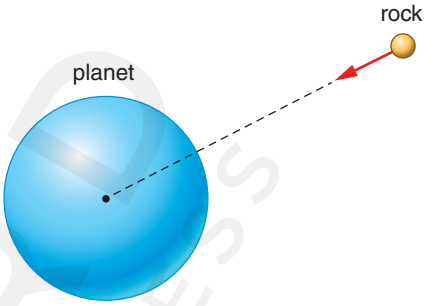


Fig a

A Its velocity is constant.

B Its acceleration is constant.

C Its acceleration is increasing.

D Its acceleration is decreasing.

***6** Which of the following statements about the gravitational field strength due to a planet is/are correct?

(1) It always points towards the centre of the planet.

(2) Its magnitude is inversely proportional to the square of the distance r from the planet's surface.

(3) Its magnitude is at its maximum on the surface of the planet.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

***7** Curiosity is a rover designed to explore Mars (Fig b). Its weight on the Earth was 8830 N. The mass of Mars is 0.107 times that of the Earth. The radius of Mars is 0.532 times that of the Earth.

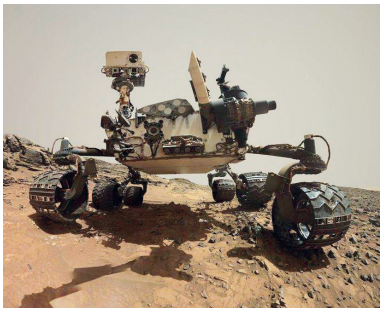


Fig b

What is the weight of Curiosity on Mars' surface?

A 503 N B 1780 N

C 3340 N D 23 400 N

Ext

8 SOHO (Fig c) is a spacecraft that was launched to study the Sun. When it is 1.48×10^8 km from the centre of the Sun, the gravitational force exerted on it by the Sun is 11.2 N. The Sun's mass is 1.99×10^{30} kg. Estimate SOHO's mass.

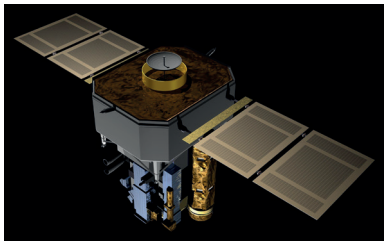


Fig c

★**9** Felix Baumgartner went skydiving from a height of 39 km above the ground (Fig d). What was his acceleration at the beginning of the dive? The Earth's radius is 6370 km.

Fig d

★**10** In the previous chapters we always take the acceleration due to gravity on Earth to be 9.81 m s^{-2}. Explain why we should not use this value for questions involving celestial bodies.

★**11** A rocket of mass 2×10^6 kg is launched from X to Y. X is on the Earth's surface while Y is $2R_E$ from the Earth's centre.

Given: Earth's mass $M_E = 5.97 \times 10^{24}$ kg, Earth's radius $R_E = 6.37 \times 10^6$ m

(a) Find the gravitational force F_X acting on the rocket when it is at X.

(b) Find the gravitational force F_Y acting on the rocket when it is at Y.

★**12** Spacecraft Voyager 1 (Fig e) was launched to study the outer solar system. Its weight was 7080 N on the Earth's surface. The Earth's radius is 6370 km. Neglect the gravitational forces from other celestial bodies.

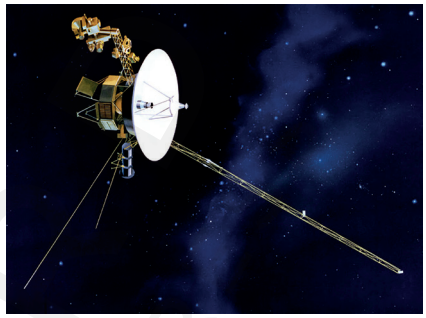


Fig e

(a) At which position would its weight be equal to 3540 N?

(b) Sketch a graph to show how its weight varies with its distance from the Earth's centre.

★**13** New Horizons (Fig f) flew by Pluto during its mission. Pluto's radius is 1150 km and the gravitational field strength at its surface is 0.658 N kg^{-1}. Estimate the gravitational field strength at the position of New Horizons when it flew 12 500 km above the surface of Pluto. Neglect the gravitational field from other celestial bodies.

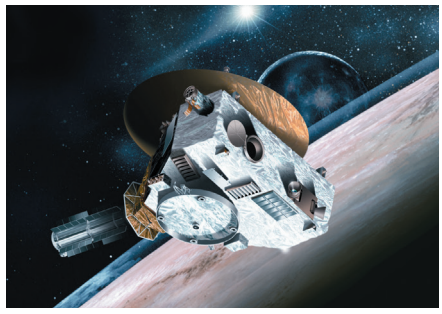


Fig f

★**14** Peter's weight is 475 N when he is on the Earth's surface. He is now on a peak which is 5895 m above the Earth's surface. The Earth's radius is 6.37×10^6 m.

(a) Find the gravitational force acting on Peter by the Earth now.

(b) If Peter releases a ball from this height, what will be the acceleration of the ball?

Ext

10.2 Circular motion under gravity

DSE goal

Determine the velocity of an object in a circular orbit.

DSE exam

12(1A)Q14, 14(1A)Q11, 14(1A)Q12, 16(1A)Q14, 18(1A)Q13, 21(1B)Q5(b), 22(1A)Q14

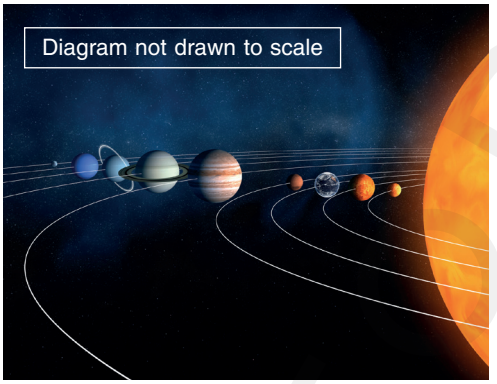
Know more

Galileo and Newton: giants who transformed the world

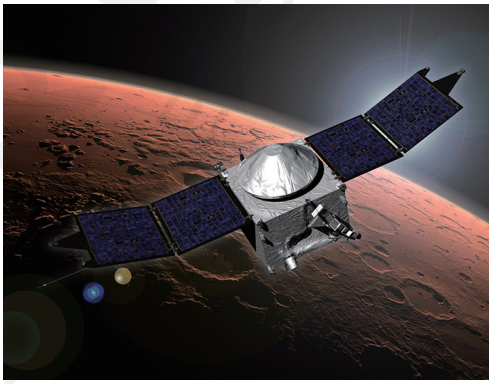
The work of Galileo and Newton revolutionized scientific thinking. Do you know what they did?

▶ p.446

Newton's law of universal gravitation, together with his three laws of motion, helps us understand the motion of planets orbiting the Sun, the paths of comets, and the motion of other celestial bodies (Fig 10.2a). These laws also help scientists fly spacecrafts towards different celestial bodies. In this unit, we will focus on circular motion under gravity.



(i) Planets orbiting the Sun.



(ii) Spacecraft flying to Mars.

Fig 10.2a *Motion of different objects in space.*

1 Gravitational force as a centripetal force

We learned in chapter 9 that a centripetal force is needed to keep an object in uniform circular motion. In space, the paths of most orbiting objects are not perfect circles (Fig 10.2b). However, we will regard all the objects as performing uniform circular motion under gravity in this part. In this case, the gravitational force provides the required centripetal force.

Diagram not drawn to scale

(i) Satellites orbiting the Earth.

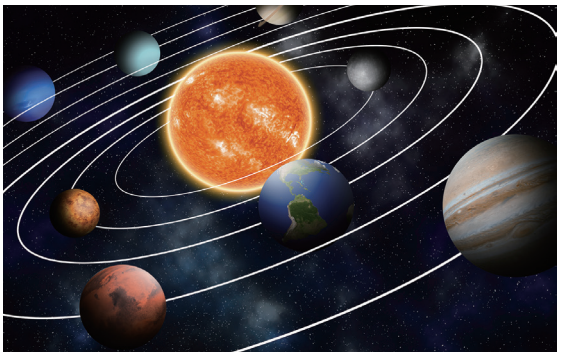


Diagram not drawn to scale

(ii) Planets in the solar system orbiting the Sun.

Fig 10.2b *Examples of orbiting objects in space.*

Ext

Consider an object of mass m revolving at a constant speed v around a massive celestial body of mass M in a circular orbit of radius r (Fig 10.2c). The object is acted on only by the gravitational force F from the celestial body.

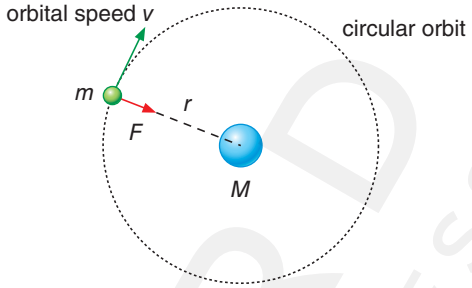


Fig 10.2c *An orbiting object performs uniform circular motion under gravity.*

As $F = \dfrac{GMm}{r^2}$ and it provides the centripetal force for uniform circular motion,

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \text{and} \quad \frac{GMm}{r^2} = mr\omega^2$$

Hence, the linear speed v and the angular speed ω of the object moving uniformly in a circular orbit are given by

universal gravitational constant = 6.67×10^{-11} N m^2 kg^{-2}

mass of celestial body that the object orbits around (kg)

$$v = \sqrt{\frac{GM}{r}}$$

linear speed of orbiting object (m s^{-1})

radius of circular orbit (m)

$$\omega = \sqrt{\frac{GM}{r^3}}$$

angular speed of orbiting object (rad s^{-1})

As the gravitational force **provides the centripetal force**, it is always perpendicular to the velocity of the orbiting object. It does **no work** on the object and does not cause any change in the speed of the object (Fig 10.2d).

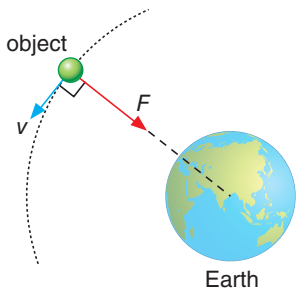


Fig 10.2d *The gravitational force acting on the object is always perpendicular to the object's velocity.*

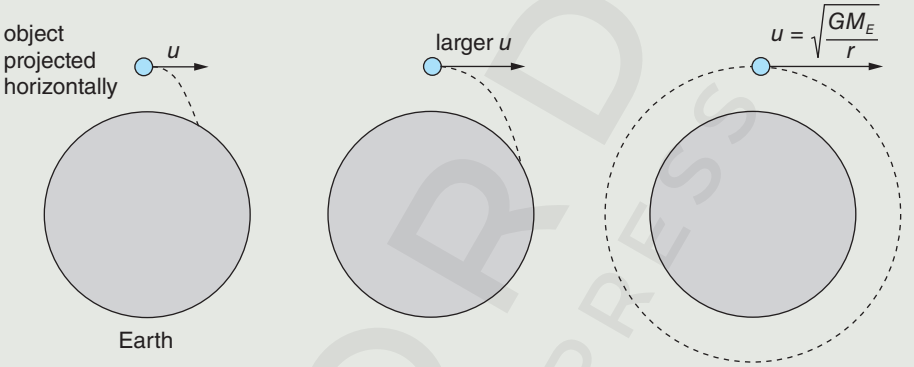
Ext

More details of orbital motion under gravity will be discussed in Book E1 Ch 2.

Know more

Orbital motion and projectile motion

Orbital motion can be considered as a special kind of projectile motion. When the launching speed equals $\sqrt{\frac{GM_E}{r}}$, the projectile will perform uniform circular motion.



① The object lands on the ground after travelling a certain distance.

② If u increases, the object will travel further before landing.

③ If u reaches $\sqrt{\frac{GM_E}{r}}$, the object will move in a circular orbit without landing.

Know more

Discovery of Neptune

In practice, an orbiting object in space may experience two or more gravitational forces acted by different celestial bodies. To learn more, read the story of the discovery of Neptune.

▶ p.450

Example 7 Orbital speed of a space station

The core module of the Tiangong space station revolves around the Earth uniformly in a circular orbit approximately 400 km above the Earth's surface. The Earth's mass and radius are 5.97×10^{24} kg and 6370 km respectively. Estimate the space station's

(a) orbital speed, and

(b) orbital period (in min).

Solution

(a) $r = (6370 + 400) \times 10^3 = 6.77 \times 10^6$ m

By $\frac{GMm}{r^2} = \frac{mv^2}{r}$,

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})}{6.77 \times 10^6}} = 7670 \text{ m s}^{-1}$$

The orbital speed of the space station is 7670 m s^{-1}.

(b) Orbital period $= \frac{2\pi r}{v} = \frac{2\pi \times (6.77 \times 10^6)}{7670} = 5546 \text{ s} = 92.4 \text{ min}$

Ext

Learn by practice 3

Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

1 A satellite revolves around the Earth in a circular orbit at 6.22×10^7 m away from the Earth's centre. The mass of the Earth is 5.97×10^{24} kg. Find the satellite's

(a) linear speed,

(b) angular speed.

2 An object moves in a circular orbit around the Earth. Its distance from the Earth surface is 1000 km. The mass and radius of the Earth is 5.97×10^{24} kg and 6370 km respectively. Find the object's

(a) linear speed,

(b) angular speed.

3 A planet moves in a circular orbit around the Sun at a constant speed of 4200 m s^{-1}. Find the radius of its orbit. Given: Sun's mass = 1.99×10^{30} kg

4 Assume that a moon moves in a circular orbit around Jupiter at a constant angular speed of 0.002 rad s^{-1}. Find the radius of its orbit. Given: Jupiter's mass = 1.90×10^{27} kg

2 Orbits around the same celestial body

Objects may take different circular orbits to **revolve around the same celestial body under gravity**. However, the centres of these orbits must be positioned at the centre of the celestial body. This is because the gravitational force acting on the object must always point towards the centre of the orbit to provide the centripetal force. Figure 10.2e shows some possible and impossible orbits.

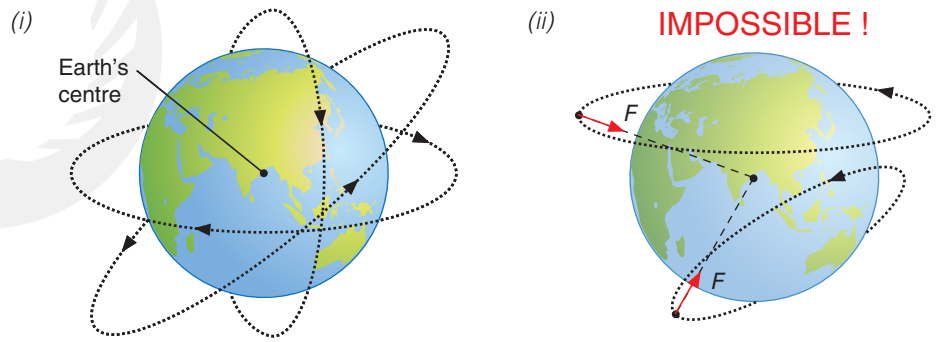


Fig 10.2e *(i) Possible orbits. (ii) Impossible orbits (a revolving object would deviate from its circular orbit due to the gravitational force).*

Ext

According to $v = \sqrt{\frac{GM}{r}}$ and $\omega = \sqrt{\frac{GM}{r^3}}$, we can conclude the following about the circular orbits around the same celestial body.

1 Objects in orbits with the same radius r have the same orbital speed v and the same orbital period T $(= \frac{2\pi}{\omega})$, regardless of their masses.

$r\uparrow \Rightarrow \omega\downarrow \Rightarrow T\uparrow$

▶ **2** Objects in a further orbit (larger r) have a lower orbital speed v and a longer orbital period T (Fig 10.2f).

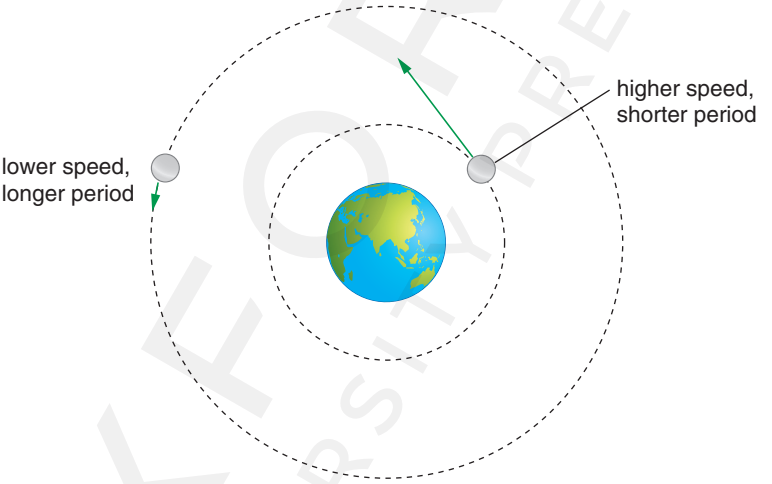


Fig 10.2f *Circular orbits of different radii.*

There is a special kind of orbit around the Earth called the *geostationary orbit*. Objects in this orbit revolve around the Earth with the same period as the Earth's rotation (24 hours), so they appear stationary to an observer on Earth. The orbit is unique; it is at a fixed height above the equator (Fig 10.2g).

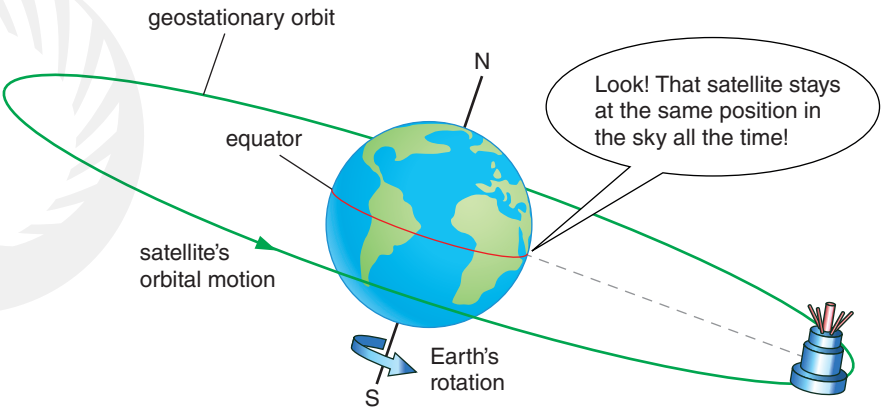


Fig 10.2g *A satellite in the geostationary orbit always appears at the same position to an observer on Earth while it is actually moving around the Earth.*

geostationary orbit 地球靜止軌道

Ext

Example 8 | Geostationary orbit

(a) By considering the gravitational field strength on the Earth's surface, show that $GM_E = 3.98 \times 10^{14}$ N m^2 kg^{-1}, where M_E is the Earth's mass. Given: Earth's radius $R_E = 6.37 \times 10^6$ m

(b) A satellite is allowed to move freely in the geostationary orbit around the Earth. It appears to always be stationary above a certain place at the equator.

(i) What is the period of the satellite?

(ii) Find the radius of the geostationary orbit.

Solution

(a) Gravitational field strength on the Earth's surface $= \dfrac{GM_E}{R_E^2}$

$= 9.81$ N kg^{-1}

$GM_E = 9.81 \times R_E^2 = 9.81 \times (6.37 \times 10^6)^2 = 3.98 \times 10^{14}$ N m^2 kg^{-1}

(b) (i) 24 hours

(ii) $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{24 \times 60 \times 60} = 7.27 \times 10^{-5}$ rad s^{-1}

By $\dfrac{GMm}{r^2} = mr\omega^2$,

$r = \sqrt[3]{\dfrac{GM}{\omega^2}} = \sqrt[3]{\dfrac{3.98 \times 10^{14}}{(7.27 \times 10^{-5})^2}} = 4.22 \times 10^7$ m

The radius of the geostationary orbit is 4.22×10^7 m.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 Which of the following statements about a satellite moving in a circular orbit around the Earth is/are correct? Put a tick in the corresponding box(es).

- ☐ The orbital speed of the satellite is independent of its mass.
- ☐ A satellite in an orbit of a smaller radius has a lower linear speed.
- ☐ A satellite in an orbit of a larger radius has a lower angular speed.
- ☐ A satellite in an orbit of a smaller radius has a shorter orbital period.
- ☐ A satellite moving in the geostationary orbit has an orbital period of 365 days.

Ext

Writing practice

Question

Objects A and B move around the Earth in circular orbits. If A is further away from the Earth than B, how is the speed of A compared to that of B? Explain your answer. (2 marks)

Answer

According to $v = \sqrt{\frac{GM}{r}}$, as A has a larger radius r than B, 1A

it has a lower speed v than B. 1A

Practise yourself

1 Objects X and Y move around a planet in circular orbits. If the radius of X's orbit is smaller than that of Y's orbit, how is the orbital period of X compared to that of Y? Explain your answer. (3 marks)

2 How is the orbital speed of a satellite related to the mass of the satellite itself and the mass of the celestial body it revolves about in a circular orbit? Explain your answer. (3 marks)

Exercise 10.2

Take $g = 9.81\ \text{m s}^{-2}$ (close to the Earth)
$G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$

1 Two satellites travel with the same angular speed around the Earth in circular orbits. Which of the following statements about the satellites must be correct?

(1) They travel in orbits at the same height above the Earth's surface.

(2) They have the same linear speed.

(3) They have the same mass.

A (2) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

***2** Two identical satellites X and Y move around the Earth in circular orbits. The gravitational force acting on X is larger than that on Y. Which of the following quantities of X is/are also larger than that of Y?

(1) Orbital period

(2) Angular speed

(3) Kinetic energy

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

Ext

*3 Dione is a moon of Saturn (Fig a). The radius of its orbit is about the same as that of the Moon (assume both orbits are circular). Given that the mass of Saturn is much larger than the Earth's mass, which of the following about the orbital period T of Dione is correct? Assume the orbital period of the Moon is T_M.

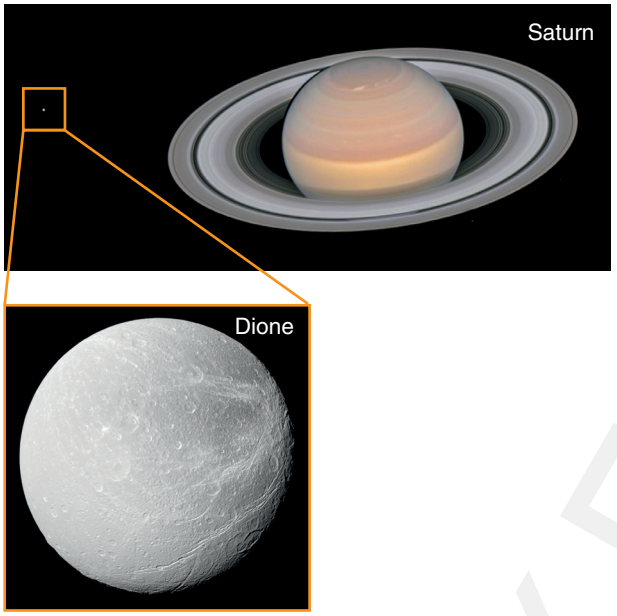


Fig a

A $T = T_M$

B $T > T_M$

C $T < T_M$

D The value of T depends on Dione's mass.

4 The figure shows a moon of a planet (Fig b). The moon revolves in a circular orbit. Draw a line to represent the plane on which the moon's orbit lies.

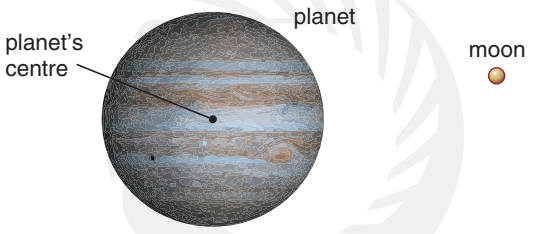


Fig b

5 The mass of the Sun is 1.99×10^{30} kg. Assume that Jupiter revolves around the Sun in a circular orbit of radius 7.78×10^{11} m. Find Jupiter's

(a) angular speed,

(b) orbital period.

6 Assume that the Earth moves around the Sun in a circular orbit once every 365 days.

Given: Sun's mass = 1.99×10^{30} kg,
Earth's mass = 5.97×10^{24} kg

(a) Find the angular speed of the Earth.

(b) Find the distance of the Earth from the Sun.

*7 Scientists record the motion of LAGEOS (Laser Geodynamics Satellite) (Fig c) to measure the value of GM_E, where M_E is the mass of the Earth. Assume that LAGEOS travels in a circular orbit of radius 1.23×10^7 m around the Earth in 3.76 hours. Estimate the value of GM_E based on this information.

Fig c

*8 Venus is a planet in the Solar System. It revolves about the Sun once every 225 days. Assume that the orbits of Venus and the Earth are both circular.

(a) Determine and explain which planet, Venus or the Earth, is closer to the Sun.

(b) Which planet, Venus or the Earth, has a higher linear speed?

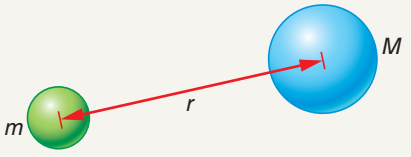
Ext

Key concepts 10

1 Newton's law of universal gravitation

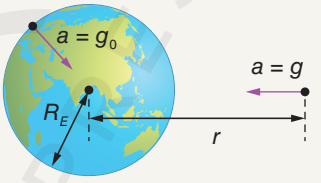
Gravitational force F between two objects:

$$F = \frac{GMm}{r^2}$$



Acceleration due to gravity g at r from Earth's centre:

$$g = \frac{GM_E}{r^2} = \frac{GM_E}{R_E^2} \times \frac{R_E^2}{r^2} = g_0 \times \frac{R_E^2}{r^2}$$



2 Gravitational field strength g

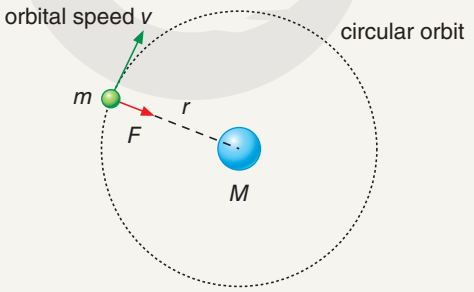
- With the same symbol and the same value as the acceleration due to gravity
- $g = \dfrac{\text{gravitational force}}{\text{test mass}} = \dfrac{F}{m} = \dfrac{GM}{r^2}$
- Vector
- Unit: N kg^{-1} (equivalent to m s^{-2})

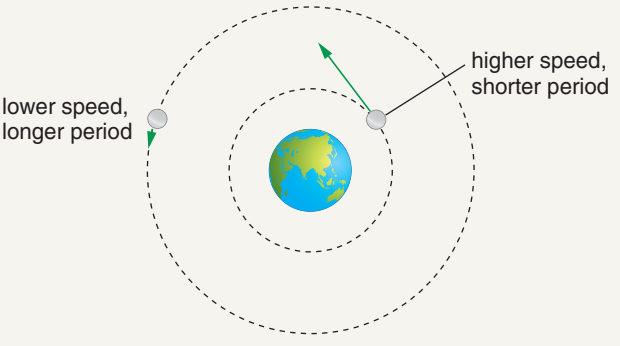
3 Circular motion under gravity

- For orbiting objects in space, the centripetal force is provided by the gravitational force

$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$$

$$\Rightarrow v = \sqrt{\frac{GM}{r}} \quad \text{and} \quad \omega = \sqrt{\frac{GM}{r^3}}$$





- Objects in the same orbit have the same orbital speed
- Object in a further orbit has a lower orbital speed and a longer orbital period
- The centre of an orbit around object X must be at the centre of X
- Orbiting objects move with constant KE

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Ext

Key terms

1 gravitational field 引力場 p.423
2 gravitational field strength 引力場強度 p.424
3 gravitational force 引力 p.415
4 Newton's law of universal gravitation 牛頓萬有引力定律 p.415
5 universal gravitational constant 萬有引力常數 p.415

Revision 10

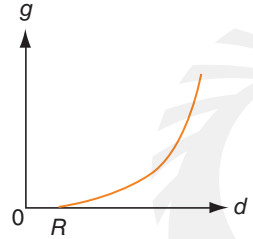
Take $g = 9.81\ \text{m s}^{-2}$ (close to the Earth)
$G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$

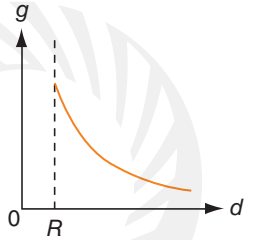
Multiple-choice questions

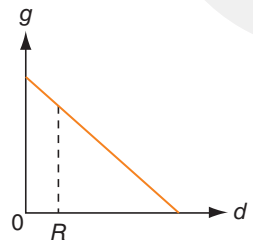
★1 10.1 The actual orbit of the Earth around the Sun is not circular. During one revolution, the largest distance of the Earth from the Sun is about 1.034 times the smallest distance. If the largest gravitational force acting on the Earth by the Sun is F, what is the smallest gravitational force?

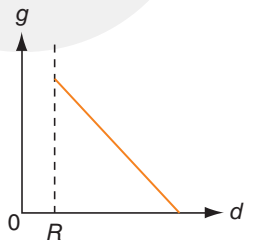
A 0.935F
B 0.967F
C 1.034F
D 1.069F

★2 10.1 Which of the following graphs best describes how the gravitational field strength g due to a planet varies with distance d from the centre of the planet? The radius of the planet is R.

A


B


C


D


★3 10.1 The acceleration due to gravity on the Moon's surface is about $\frac{1}{6}$ of that on the Earth's surface. The mass of the Moon is about $\frac{1}{81}$ of that of the Earth. The ratio of the Moon's radius to the Earth's radius is about

A 1 : 4.
B 1 : 14.
C 1 : 18.
D 1 : 22.

★4 10.2 Two satellites P and Q move around the Earth in circular orbits. They have the same mass but the orbit of P has a larger radius. Which of the following statements about P and Q is/are correct?

(1) P has a shorter period than Q.
(2) P has a smaller acceleration towards the Earth's centre than Q.
(3) P has a lower KE than Q.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

★5 10.2 Two satellites P and Q move around the Earth in circular orbits. Their orbits are of the same radius but P has a smaller mass. Which of the following statements about P and Q is/are correct?

(1) P has a shorter period than Q.
(2) P has a smaller acceleration towards the Earth's centre than Q.
(3) P has a lower KE than Q.

A (3) only
B (1) and (2) only
C (1) and (3) only
D (2) and (3) only

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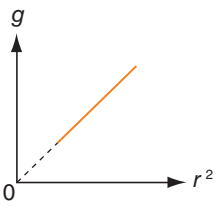
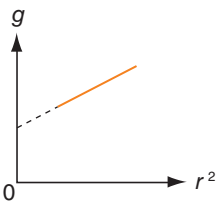
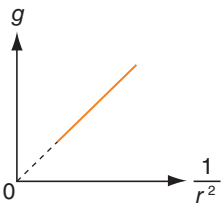
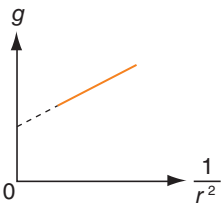
Ext

***6** 10.1 Which of the following graphs best describes the relationship between the gravitational field strength g due to a planet and the distance r from the centre of the planet?

A 

B 

C 

D 

***7** 10.2 Two objects revolve around a planet in circular orbits. Which of the following statements about the two objects is/are correct?

(1) If their orbital radii are the same, the centripetal forces they need must be the same.

(2) If their orbital radii are the same, their orbital periods must be the same.

(3) Their orbits must lie on the same plane.

A (2) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

***8** 10.2 Mars has two moons, Phobos and Deimos. The orbital speeds of Phobos and Deimos are v_P and v_D respectively. If the radius of Phobos' orbit is R, what is the radius of Deimos' orbit?

A $\dfrac{Rv_P^2}{v_D^2}$ B $\dfrac{Rv_D^2}{v_P^2}$

C $R\sqrt{\dfrac{v_P}{v_D}}$ D $R\sqrt{\dfrac{v_D}{v_P}}$

***9** 10.1 The radius of the Earth is R_E. Positions X and Y are R_E and $2R_E$ above the Earth's surface respectively. Find the ratio of the magnitude of the gravitational field strength at X to that at Y.

A 4 : 1

B 4 : 9

C 9 : 1

D 9 : 4

***10** 10.2 A satellite revolves around the Earth in a circular orbit. Which of the following statements about the gravitational force acting on the satellite is/are correct?

(1) It is balanced by the centripetal force of the satellite.

(2) Its magnitude depends on the mass of the satellite.

(3) It has the same direction as the satellite's velocity.

A (1) only

B (2) only

C (3) only

D (1) and (2) only

****11** 10.1 Consider objects placed on the surface of a planet. The graph below shows the relationship between their masses and their weights (Fig a).

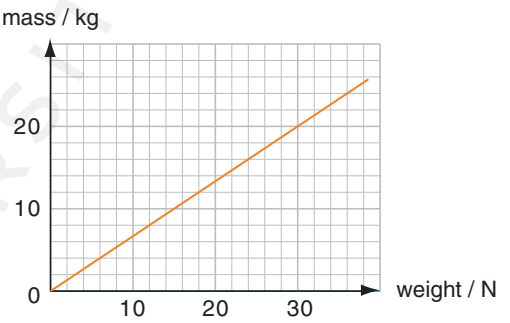


Fig a

What is the mass of the planet if its radius is 2000 km?

A 4.49×10^{16} kg

B 3.99×10^{22} kg

C 5.99×10^{22} kg

D 9.00×10^{22} kg

▶ Refer p.419

****12** 10.2 Imagine that a satellite moves in a circular orbit around the Earth very close to the Earth's surface, i.e. at about 6370 km from the Earth's centre. Which of the following about the orbital period of the satellite is correct?

A It is about 1 day.

B It is much longer than 1 day.

C It is much shorter than 1 day.

D Its value depends on the satellite's mass.

▶ Refer p.434, 435

Q15 Exam Report: Nearly a quarter of the candidates chose option B, possibly assuming that the acceleration of the object was constant throughout its fall.

Ext

10.2 13 HKDSE 2012 Paper 1A Q14

Two satellites move in circular orbits of the same radius R around the Earth (mass M). The orbits are in two different planes P and Q as shown. Plane P coincides with the Earth's equator while plane Q is inclined to the equator at θ. Which statement is **incorrect**?

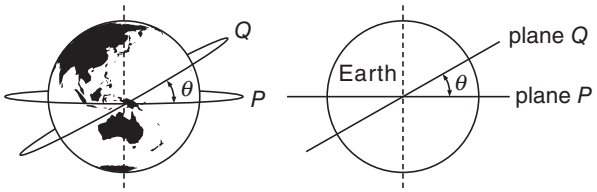


Fig b

A The speed of satellite P is $\sqrt{\frac{GM}{R}}$. (8%)

B The centripetal force acting on satellite Q is pointing along the plane Q. (16%)

C The acceleration of both satellites is the same in magnitude. (18%)

D The period of satellite Q is longer than that of satellite P. (58%)

Misc 14 HKDSE 2014 Paper 1A Q11

An astronaut inside a spacecraft moving in a circular orbit around the Earth is apparently weightless because

A the astronaut is too far from the Earth to feel the Earth's gravitational force. (25%)

B the astronaut and the spacecraft are both moving with the same acceleration towards the Earth. (29%)

C the Earth's gravitational force on the astronaut is balanced by the reaction force of the spacecraft's floor. (9%)

D the Earth's gravitational force on the astronaut is balanced by the centripetal force. (37%)

Q14 Exam Report: Over one third of the candidates attributed the astronaut's weightlessness to the Earth's gravitational force being balanced by the centripetal force.

Misc 15 HKDSE 2017 Paper 1A Q13

A small object is released from rest at a point very far away from a planet X. The object then starts moving towards X. X does not have an atmosphere. Neglect the effect of other celestial bodies.

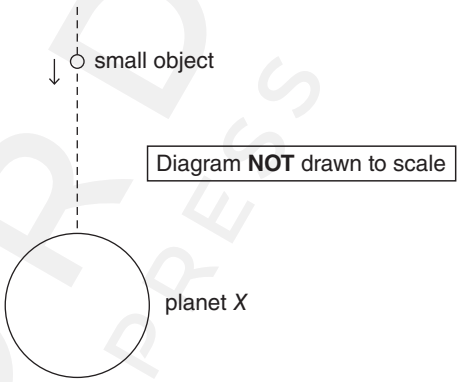
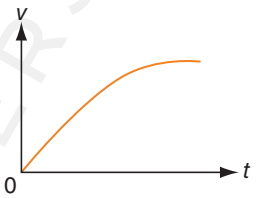


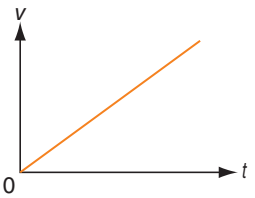
Fig c

Which of the following graphs best shows the variation of the velocity v of the object with time t before it hits X?

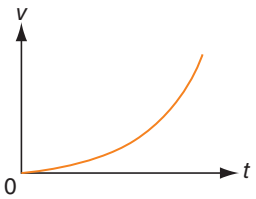
A (12%)



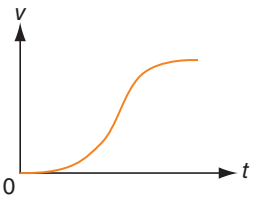
B (24%)



C (59%)



D (5%)



10.1 16 HKDSE 2020 Paper 1A Q10

The diameter of Neptune is about 4 times that of the Earth and its mass is about 17 times that of the Earth. Estimate the acceleration due to gravity on Neptune's surface.

Given: acceleration due to gravity on Earth's surface $g = 9.81\ \text{m s}^{-2}$

A $2.3\ \text{m s}^{-2}$ B $9.2\ \text{m s}^{-2}$

C $10.4\ \text{m s}^{-2}$ (60%) D $41.7\ \text{m s}^{-2}$

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Ext

Conventional questions

***17** Misc Mars is about 2.28×10^8 km from the Sun. The mass of the Sun is 1.99×10^{30} kg. Assume that Mars revolves around the Sun in a circular orbit. Neglect the effect of other celestial bodies. Estimate

(a) the gravitational field strength at the position of Mars due to the Sun, (2 marks)

(b) the orbital speed of Mars, (2 marks)

(c) the orbital period of Mars (in days). (2 marks)

***18** 10.1 During the Apollo 15 mission, astronaut David Scott, who was on the Moon's surface, dropped a hammer and a feather at the same time from the same height.

(a) Explain why the hammer and the feather reached the ground at the same time. (3 marks)

(b) Suppose that the two objects were released at a height of 1.2 m. The Moon's mass and radius are 7.35×10^{22} kg and 1740 km respectively. Find the time they take to fall to the ground. (3 marks)

***19** 10.1 In the following set-up (Fig d), balls X and Y have a great difference in mass, each suspended by a string. They are placed close to each other near the Earth's surface. Victor thinks that due to the gravitational force acted by X, Y is deflected by an angle θ which can be measured by a protractor. Explain briefly whether Victor is correct. (4 marks)

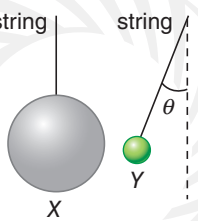

Fig d

***20** 10.2 An artificial satellite of mass m revolves around a planet in a circular orbit and experiences the gravitational force from the planet.

(a) Explain why the work done on the satellite by the gravitational force is zero. (1 mark)

(b) During a mission, the orbital speed v of the satellite is 8 km s^{-1} and its distance r from the planet's centre is 10 000 km. Estimate the mass of the planet. (2 marks)

(c) Figure e shows the graph of v against r.

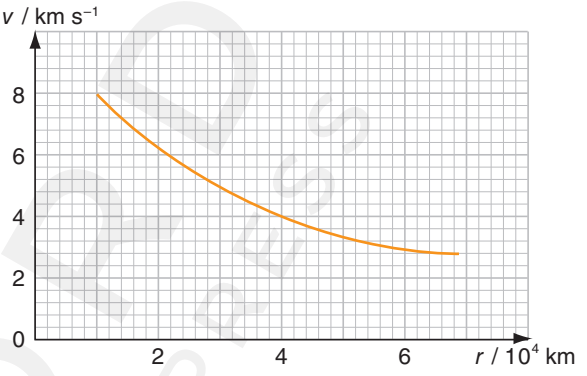

Fig e

Suppose that another satellite of mass $0.25m$ is used. Draw its graph in Figure e. (1 mark)

***21** 10.2 A satellite of mass 1000 kg revolves around the Earth in a circular orbit as shown (Fig f).

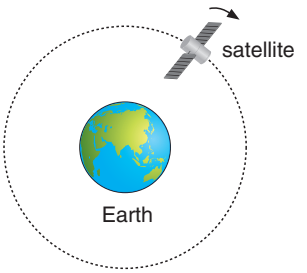

Fig f

(a) (i) Draw and label an arrow as v in Figure f to show the direction of the velocity of the satellite at the moment shown. (1 mark)

(ii) Draw and label an arrow as F in Figure f to show the direction of the net force acting on the satellite at the moment shown. (1 mark)

(b) The satellite is in an orbit 7000 km above the Earth's surface. The Earth's radius is 6370 km.

(i) Find the linear speed of the satellite. (3 marks)

(ii) Find the weight of the satellite in its orbit. (2 marks)

Ext

★22 10.1 (a) Define gravitational field strength. (1 mark)

(b) The mass of Mars is about $\frac{1}{10}$ of that of the Earth, and its radius is about $\frac{1}{2}$ of that of the Earth. Estimate the gravitational field strength on the surface of Mars in terms of g_E, the gravitational field strength on the Earth's surface. (2 marks)

(c) Opportunity (Fig g) is a rover designed to explore Mars.

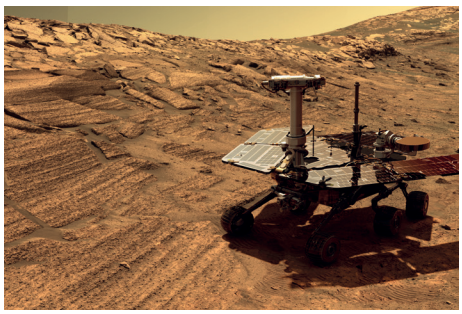

Fig g

(i) If Opportunity weighed 1810 N on the Earth's surface, find its weight on Mars' surface. (2 marks)

(ii) Suppose that when Opportunity landed on Mars, it descended vertically at a constant speed. Sketch a graph to show how the gravitational force exerted on it by Mars varied with time during its landing. (2 marks)

★23 Misc A stone is projected vertically upwards with an initial speed of 3 m s^{-1} from the surface of a planet. The stone falls back to the surface after 1.6 s. There is no atmosphere on the planet. The radius of the planet is 2300 km.

(a) Estimate the acceleration due to gravity near the surface of the planet. (2 marks)

(b) A moon moves around the planet in a circular orbit of radius 10 200 km.

(i) Find the acceleration of the moon. (3 marks)

(ii) Find the linear speed of the moon. (2 marks)

★24 10.2 (a) Show that for satellites revolving around the Earth in circular orbits, their linear speed v and angular speed ω is related by $\frac{v^3}{\omega} = GM_E$, where M_E is the radius of the Earth. (2 marks)

(b) A satellite revolves around the Earth in a circular orbit with an angular speed of 7.29×10^{-5} rad s^{-1} and a linear speed of 3.07 km s^{-1}. If another satellite revolves around the Earth in a circular orbit with a linear speed of 1.80 km s^{-1}, find its orbital period. (3 marks)

★25 10.1 The weight of an object on the surface of a planet is 200 N. When it is raised to a certain height h, its weight becomes 100 N. The radius of the planet is 3400 km.

(a) Find h. (2 marks)

(b) When the object is thrown vertically upwards from the planet's surface with an initial speed of 3 m s^{-1}, it reaches a maximum height of 0.9 m.

(i) Explain why we can take the acceleration of the object during its upward motion as constant. (2 marks)

(ii) Estimate the mass of the object. (3 marks)

(c) Find the mass of the planet. (2 marks)

★26 10.2 Ganymede and Callisto are two moons of Jupiter revolving in circular orbits. The radii of Ganymede and Callisto are 1.07×10^9 m and 1.88×10^9 m respectively. The gravitational field strength at the orbit of Callisto due to Jupiter is 0.0357 N kg^{-1}.

(a) Find the orbital speed of Ganymede. (3 marks)

(b) A rock of mass 100 kg revolves around Jupiter in Ganymede's orbit. Find the gravitational force acting on the rock by Jupiter. (2 marks)

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Ext

***27** 10.1 Kelvin performs the Cavendish experiment (see Know more on p.445) to determine the value of G. Figures h and i show schematically the structure of the experimental set-up.

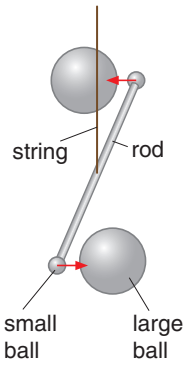


Fig h

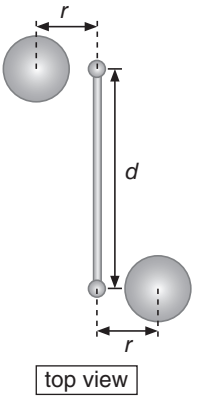


Fig i

Two small metal balls are connected by a rod which is suspended by a string. A large metal ball is placed near each small ball. The gravitational attraction between the balls turns the rod. By knowing the turning angle of the rod, the moment of force acting on the rod can be calculated. Then this value can be used to find the value of G.

Kelvin records his measurement as follows.

Distance between small balls (d) = 10 cm

Mass of each small ball (m) = 50 g

Mass of each large ball (M) = 1500 g

Distance between large ball and small ball (r) = 4.5 cm

Moment of force acting on the rod (τ) = 2.1×10^{-10} N m

(a) Find the value of G according to the above data. (3 marks)

(b) It is given that the accepted value of G is 6.67×10^{-11} N m^2 kg^{-2}. Find the percentage error of the answer to (a). 15.0% (1 mark)

(c) Besides the error in making measurements, suggest another source of error in this experiment. (1 mark)

***28** 10.2 (a) An object revolves around a planet of mass M in a circular orbit of radius r with a period T. Show that $T^2 = \dfrac{4\pi^2 r^3}{GM}$. (2 marks)

(b) Saturn is easily recognized because of its rings as shown (Fig j). The rings that spread out to form a wide belt are composed of ice, rocks and dust which orbit Saturn in circular paths.

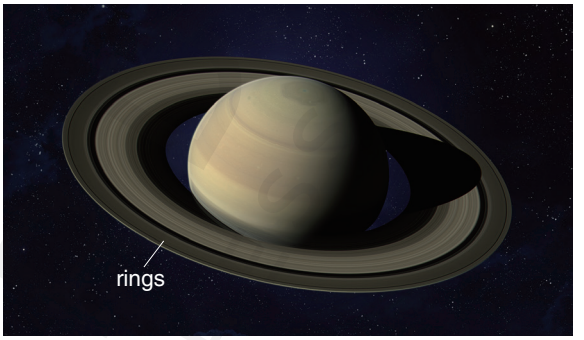


Fig j

(i) Where is the centre of the rings? (1 mark)

(ii) Do all the rocks in the rings revolve around Saturn with the same period? Explain. (2 marks)

****29** 10.2 The orbital period T and orbital radius r of the four largest moons of Jupiter are shown in Table a.

Moon of Jupiter	T / days	r / km
Io	1.79	422 000
Europa	3.55	671 000
Ganymede	7.16	1 070 000
Callisto	16.7	1 880 000

Table a

(a) Plotting T against r^n would give a straight-line graph.

(i) What is the value of n? (1 mark)

(ii) Plot the graph using the data in Table a. (2 marks)

(b) Use the graph in (a) to estimate the mass of Jupiter. (2 marks)

(c) Metis is the innermost moon of Jupiter. Its orbital radius is 128 000 km. Estimate its orbital speed. (2 mark)

▶ Refer p.434

Ext

★★30 Misc

A satellite moves around the Earth in a circular orbit 900 km above the Earth's surface. The Earth's radius is 6370 km.

(a) Find the gravitational field strength at the orbit of the satellite due to the Earth. (2 marks)

(b) Find the period of the satellite. (3 marks)

(c) If the satellite is repositioned to a higher orbit, how will its period change? Explain briefly. (2 marks)

Refer p.425, 426

31 HKDSE 2021 Paper 1B Q5

Misc

A rocket carrying an artificial satellite is launched vertically from the Earth. When the rocket is at a certain height from the Earth's surface, it expels 2.60×10^3 kg of gas per second with a certain speed v towards the Earth's centre. As a result, an average thrust of 5.20×10^6 N is produced. Neglect air resistance.

(a) (i) Assuming that the speed of the rocket is negligible, estimate v. (2 marks)

(ii) At a certain instant, the total mass of the rocket and the artificial satellite is 3.60×10^5 kg while the acceleration due to gravity at the rocket's position is 8.56 m s^{-2}. Estimate the acceleration a of the rocket at this position. (2 marks)

(iii) Suppose the rocket keeps expelling gas at the same rate for a few seconds. Would the rocket's acceleration increase, decrease or remain unchanged in that duration? Explain. (2 marks)

(b) The artificial satellite is put in the geostationary orbit of radius r around the Earth. It appears to be always stationary above an observer at the equator.

(i) State the period of this satellite. (1 mark)

(ii) Show that r is approximately 42 000 km. (g = 9.81 m s^{-2})

Given: radius of the Earth = 6.37×10^6 m (2 marks)

Q31 Exam Report: Candidates' performance was fair in general.

(a) (i) Some candidates failed to relate the average thrust to the rate of change of momentum of the exhaust gas.

(ii) In applying Newton's laws of motion to find the rocket's acceleration, weaker candidates omitted the weight mg or wrongly took the net force as $F_{thrust} + mg$.

(iii) Candidates performed poorly. Not many were able to point out that the average thrust remained the same while the rocket's mass decreased.

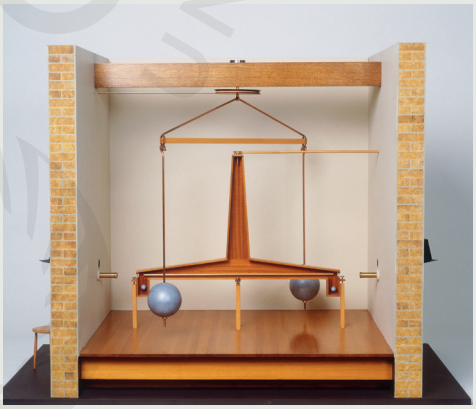
(b) (i) A few candidates mistook the period of the satellite around the Earth as 1 year or 365 days.

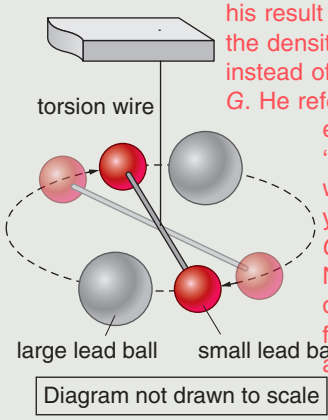
(ii) Only the more able ones knew how to make use of $\frac{GM}{R^2} = g$ to tackle this part.

Know more

The value of G (p.415)

The value of G can be found experimentally through a set-up similar to the one shown below. It works by measuring the tiny gravitational forces acting on the two small lead balls due to the two large lead balls.





The small lead balls move towards the large ones due to gravitational forces.

Cavendish expressed his result in terms of the density of the Earth instead of the value of G. He referred to his experiment as 'weighing the world'. His data yielded a value of G of 6.74×10^{-11} N m^2 kg^{-2}, which differs by only 1% from the currently accepted value.

The first measurement was made by *Henry Cavendish* in 1798, long after the time of Newton.

You may watch the following video to learn more about the experiment.

Video

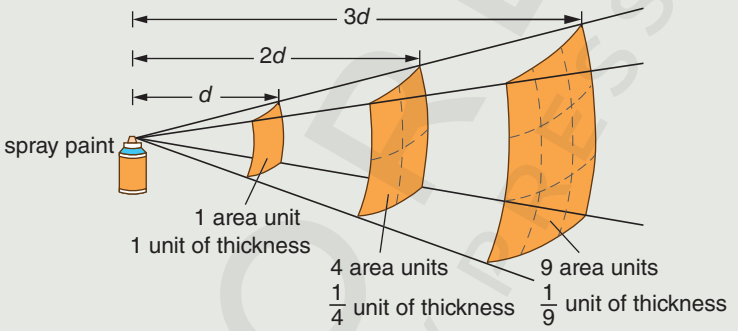
Henry Cavendish 卡文狄什

Ext

Know more

Inverse-square law (p.418)

The inverse-square law can be understood by thinking about how paint from a can of spray paint spreads with increasing distance r. Suppose we spray paint 1 m from a wall and produce a square patch of paint of 1 mm thick. How thick would the patch be if we spray the same amount of paint 2 m from the wall? The paint would spread over an area 4 times as big, and its thickness would become $\frac{1}{4}$ mm. Hence thickness of paint $\propto \frac{1}{r^2}$.



Know more

Galileo and Newton: giants who transformed the world (p.430)

For many centuries in the western world, people had been heavily influenced by thinkers of ancient times such as Aristotle. They generally thought that:

A force is needed to keep an object moving.

Heavier objects fall faster to the ground than lighter objects.

Motion on the Earth and motion in the heavens follow different sets of natural laws.

The Earth is the centre of the universe. The Sun, the Moon and planets move in perfect circles around the Earth.

Galileo (1564–1642) was the first person to test old ideas about motion through careful observations and by doing experiments (actually, some were carried out in his mind).

All objects fall at the same rate, regardless of their weight.

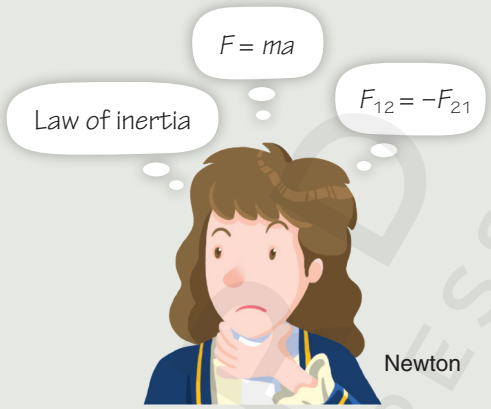
Objects tend to move at a constant speed along a straight line.

Galileo

Galileo also observed the day and night sky with the newly invented telescope. Read Book E1 chapter 1 for details.

Ext

Galileo died in 1642. In that same year, Isaac Newton (1642–1727) was born. Based on Galileo's findings, Newton formulated his three laws of motion.



Newton also devised his theory of universal gravitation through which the motion of planets, comets, stars and galaxies can be understood.

Even more, Newton invented a mathematical language that we now call 'calculus' and made significant contributions to our understanding of light.

Newton's work completely refuted the view that the Earth is the centre of universe. He firmly established that the Earth is one of the many planets that move around the Sun.

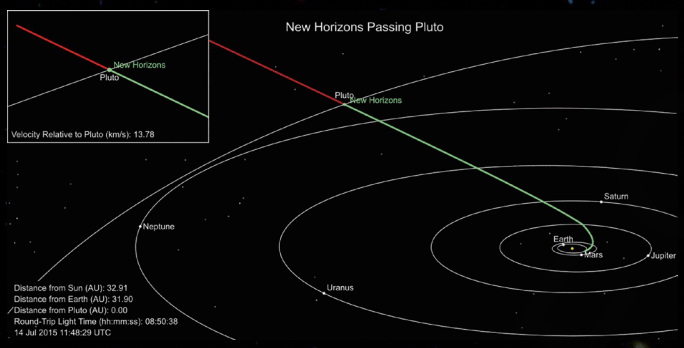
Newton showed that motions on the Earth and motions in the heavens are governed by the same set of natural laws.

Galileo and Newton established the importance of observations and the use of experiments and mathematics in studying the natural world. They opened the door to modern science. They discovered natural laws and by doing so shared with us the conviction that the physical world is comprehensible. They completely changed how people perceive the world and how we study the world.

Science and technology and

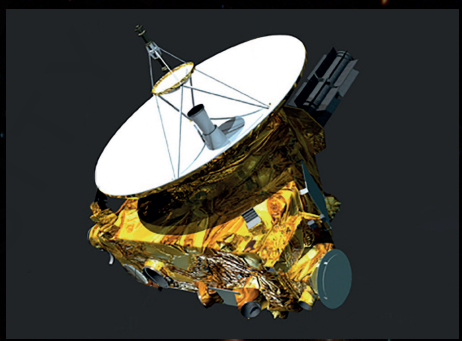
Our knowledge of space grew quickly thanks to developments in science and technology. Below are some recent examples.

2015 The spacecraft New Horizons flew by the dwarf planet Pluto. After a decade-long journey, it travelled 7.5 billion km and flew by the planet within 12 500 km to make a survey.



Journey of New Horizons.

First close-up photograph of Pluto.

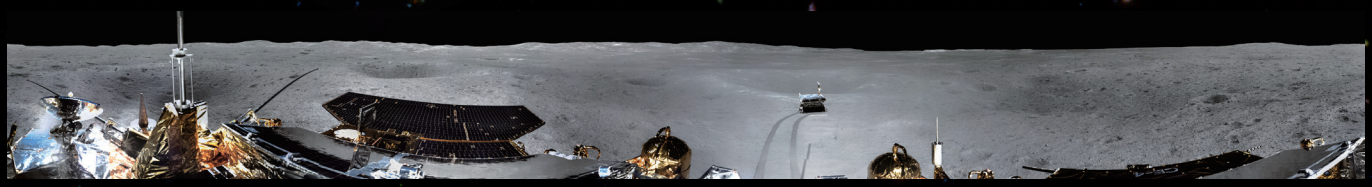


New Horizons spacecraft.

2016 The LIGO and Virgo scientific collaboration announced the first detection of gravitational waves. The two perpendicular arms of the detector are each of length 4 km.

A structure that detects gravitational waves.

2019 A lunar rover (Yutu-2) landed on the far side of the Moon for the first time. Signals from the far side of the Moon cannot reach the Earth directly. They have to be relayed to a satellite orbiting the Moon.



The first panorama view of the far side of the Moon.

the exploration of outer space

2019 Not all telescopes detect visible light. Some telescopes are built to detect radio signals from space. The largest one, called FAST (Five-hundred-metre Aperture Spherical radio Telescope), started operation in 2019.

FAST has a diameter of 500 m.

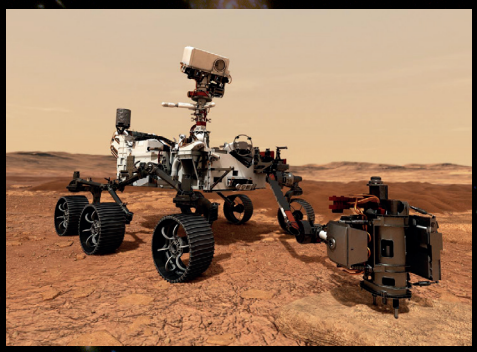
2019 The first image of a black hole was taken using a network of many telescopes in different places on the Earth.

The first image of a black hole.

One of the arrays of telescopes.

The arrays of telescopes are set up in different places on the Earth.

2020 The first Mars sample return mission was launched. Rocks and soil samples from Mars will be collected and returned to Earth for analysis. The results will help us answer some key questions about the potential for life on Mars.



Perseverance rover collecting samples on Mars.

The range of tide varies everyday. The tide's range is maximum during new or full moon when the Sun, Earth and Moon are aligned. This is called the spring tide. When the Moon is at the first or third quarter, the tide's range is minimum and is called neap tide.

Ext

Know more

Tides (p.416)

In Sai Kung, there is an island called *Kiu Tsui Chau*. At low tide, people are able to reach the island on foot from the *opposite bank*, but the path is submerged in water at high tide (Fig a). Tides are mainly caused by the **difference** in the gravitational pull by the Moon on the opposite sides of the Earth (Fig b).

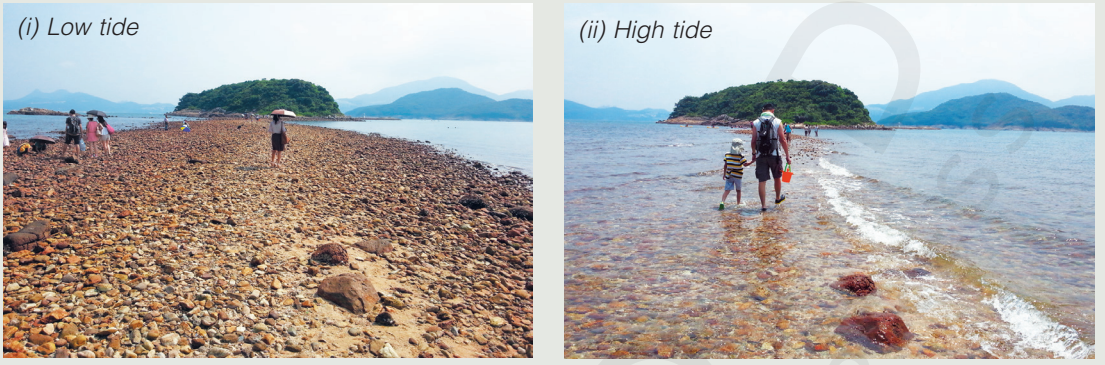


Fig a

The Sun's gravitational pull on the Earth is about 180 times greater than the Moon's. The Sun's pull can also produce a tidal effect, but it is weaker than that produced by the Moon. This is because the difference in gravitational pull on the opposite sides of the Earth by the Sun is only about half of that by the Moon. You may verify this with the following data.

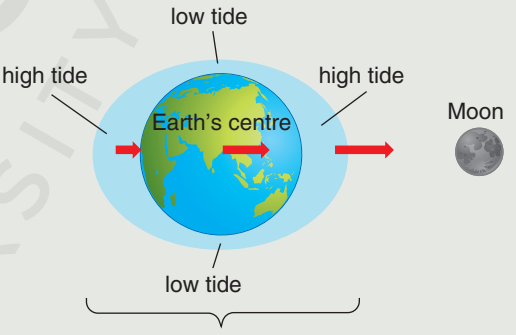
Sun's mass = 1.99×10^{30} kg

Moon's mass = 7.35×10^{22} kg

Distance of Sun from Earth = 1.50×10^{11} m

Distance of Moon from Earth = 3.84×10^{8} m

Earth's radius = 6370 km



The ocean is elongated due to the difference in the gravitational pulls on the near side and far side of the Earth.

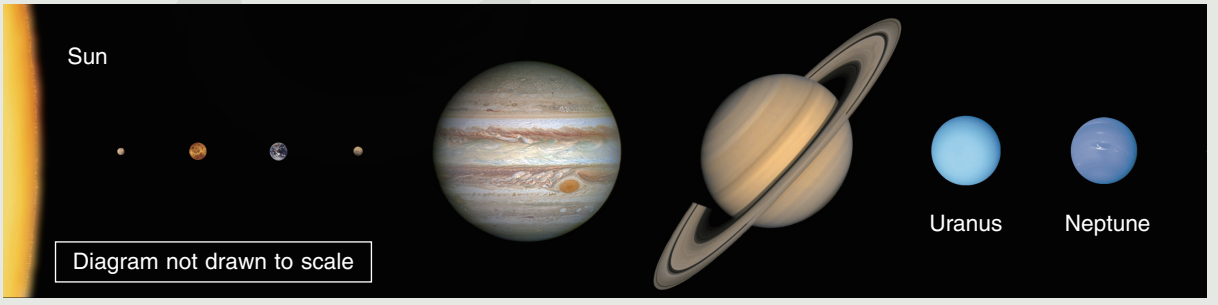
Fig b

OUP eResources

Know more

Discovery of Neptune (p.432)

By the 1840s, scientists had found that the orbit of Uranus, the furthest planet known at that time, deviated from the orbit predicted using Newton's laws. By using Newton's law of gravitation, two astronomers *Urbain Le Verrier* and *John Couch Adams* predicted independently that there must be an unknown neighbouring planet applying a gravitational force to disturb the orbit of Uranus. From their prediction, the planet Neptune was eventually discovered in 1846, 159 years after Newton published his law of gravitation.



Uranus and Neptune are the 7th and 8th major planets from the Sun (not in scale).

Kiu Tsui Chau 橋咀洲　opposite bank 對岸　Urbain Le Verrier 勒維耶
John Couch Adams 阿當斯